

Lecture notes: Lang's FCC and Jacobians

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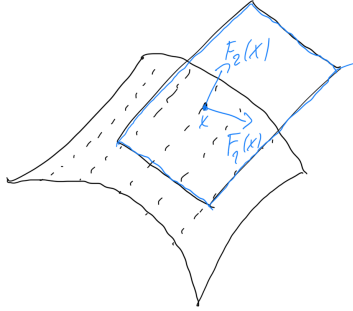
Abstract

We will go over the basic ideas of the paper [5], mainly concerning the passage from the "distributional" problem concerning the metric currents to the "functional" one concerning Jacobians of Lipschitz functions. Slightly less detail will be provided for showing the lack of regularity for the Jacobian problem.

1 Currents and Jacobians

1.1 Intro to Lang's currents

Goal: Encode information about a surface into only metric structure of $\mathbb{R}^d$.



$$\langle M \rangle(f, \pi^1, \pi^2) = \int_M f \det\left(\frac{\partial \pi^j}{\partial F_i}\right)_{i,j=1}^2 d\mathcal{H}^2$$

Makes sense for Lipschitz scalars f, π^1, π^2 by Rademacher!

$$(L) \quad \pi^1 \text{ or } \pi^2 \text{ const. on spt } f \implies \langle M \rangle(f, \pi^1, \pi^2) = 0$$

$$(JC) \quad f_i, \pi_i^1, \pi_i^2 \rightarrow f, \pi^1, \pi^2 \text{ pointwise with uniformly bounded Lipschitz constants} \implies \langle M \rangle(f_i, \pi_i^1, \pi_i^2) \rightarrow \langle M \rangle(f, \pi^1, \pi^2)$$

Definition 1.1 (Metric currents). Suppose X is a compact metric space. A metric k -current on X is a multilinear map

$$T: \text{Lip}(X) \times \text{Lip}(X)^k \rightarrow \mathbb{R} \quad (1.1)$$

(elements of $\text{Lip}(X)$ are denoted by $\pi = (\pi^1, \dots, \pi^k)$) satisfying

(L) If there is some $j \in \{1, \dots, k\}$ such that $\pi_j = 0$ on a neighbourhood of $\text{spt } f$, then $T(f, \pi) = 0$.

(JC) If each π_i^j converges to π^j and f_i converges to f (pointwise with bounded Lipschitz constants), then

$$T(f_i, \pi_i) \rightarrow T(f, \pi). \quad (1.2)$$

Space of all metric k -currents on X is denoted by $\mathcal{D}_d = \mathcal{D}_d(X)$.

Remark 1.2. For compact metric spaces the definition provided is equivalent to that of Lang [4]. To recover the definition of Ambrosio and Kirchheim [1], one needs to add the axiom of finite mass, which we will not discuss in detail. If X is merely locally compact or even just complete, the definition may be adjusted to yield similar theory in this more general setting.

Conjecture 1.3. Every metric k -current in $\mathbb{R}^d$ supported inside a compact set is a flat k -chain in the sense of Federer-Fleming.

Simplification in the top dimension Let $Q = [0, 1]^d \subset \mathbb{R}^d$ and suppose $u \in L^1(Q)$. We define

$$\text{Em}_*(u): \text{Lip}(Q) \times \text{Lip}(Q)^d \rightarrow \mathbb{R} \quad (1.3)$$

by

$$\text{Em}_*(u)(f, \pi) = \int_Q u f \det D\pi d\mathcal{L}^d. \quad (1.4)$$

This *always* defines a metric d -current.

Conjecture 1.4 (Lang's conjecture for top currents). *For every metric d -current T in Q there is some $u \in L^1(Q)$ such that $T = \text{Em}_* u$. In other words, $\text{Em}_*: L^1 \rightarrow \mathcal{D}_d$ is surjective.*

Now we will try to understand more about the operator Em_* by studying the dual operator. To do that we need to develop a theory of “potential forms” - objects on which metric currents act.

1.2 Metric compactum, normability of currents

Preliminaries Let X be a metric space with a distinguished point $0 \in X$.

$$\text{Lip}_0(X) = \{f: X \rightarrow \mathbb{R} \text{ Lipschitz}, f(0) = 0\} \quad (1.5)$$

is normed with $\text{Lip}(\cdot)$.

$$\text{Lip}_b(X) = \{f: X \rightarrow \mathbb{R} \text{ Lipschitz and bounded}\} \quad (1.6)$$

is normed with $\|\cdot\|_L := \max\{\text{Lip}(\cdot), \|\cdot\|_\infty\}$. Note that if X is compact then $\text{Lip}_b(X)$ coincides with the space of all Lipschitz maps $\text{Lip}(X)$.

Proposition 1.5. *Both spaces above are dual Banach spaces. If X is separable, then the preduals are separable. The weak* convergence on both of the unit balls is pointwise convergence. If X is compact, then the weak* convergence on both of the balls is metrized by*

$$d(f, g) = \|f - g\|_\infty, \quad (1.7)$$

so in particular, the unit balls equipped with this metric are compact (cf. Arzela-Ascoli).

Metric compactum From now on, X is a compact metric space.

Definition 1.6 (Metric compactum). For a compact metric space X we define the *metric k -compactum*

$$\mathcal{K}_k = (B_{\text{Lip}_b(X)}, w^*) \times (B_{\text{Lip}_0(X)}, w^*)^k \quad (1.8)$$

equipped with the (compact) product topology.

Note that by (JC) every metric k -current on X is an element of $C(\mathcal{K}_k)$.

Norm on the space of metric k -currents We let

$$\|T\|_{\mathbb{P}_d} = \|T\|_{\mathbb{P}_d(X)} = \|T\|_{C(\mathcal{K}_k)} \quad (\text{the latter being the supremum norm})$$

The norm is called the *potential norm*, currents for which it is finite are called potential currents. If X is compact, all metric currents are potential. The space of all potential currents is denoted by $\mathbb{P}_k$.

Remark 1.7. If X is only locally compact, similar theory makes sense but not every metric current is potential. Potential currents are those metric currents that “do not explode at the boundary/infinity”. This justifies renaming metric to potential in this setting.

Theorem 1.8 (Defining property of potential norm). *Suppose T is a metric current and $(f, \pi) \in \text{Lip}_b(X) \times \text{Lip}_0(X)^k$. Then*

$$|T(f, \pi)| \leq \|T\|_{\mathbb{P}_k} \max\{\text{Lip}(f), \|f\|_\infty\} \prod_{j=1}^k \text{Lip}(\pi^j). \quad (1.9)$$

Proof. If $(f, \pi) \in \mathcal{K}_k$, then it follows by definition. The general case follows from multilinearity, in particular the homogeneity in each variable. $\square$

Lemma 1.9. *The space $\mathbb{P}_k$ is complete.*

Proof. It suffices to show that it is a closed subspace of $C(\mathcal{K}_k)$. Suppose T_i is a sequence of metric k -currents converging to T in $C(\mathcal{K}_k)$. By definition, the limit is then jointly continuous. Checking that it is local and multilinear is a near trivial exercise. $\square$

1.3 Potential forms

We now specialize to $X = Q = [0, 1]^d$ and to d -currents and d -forms.

Formal Lipschitz sums A *formal Lipschitz sum* is an expression of the form

$$\sum_i (f_i, \pi_i), \quad (1.10)$$

where each f_i is a Lipschitz map on Q , each $\pi_i = (\pi_i^1, \dots, \pi_i^d)$ is Lipschitz with $\pi_i(0) = 0$ and such that

$$s(\sum_i (f_i, \pi_i)) = \sum_i \max\{\text{Lip}(f_i), \|f_i\|_\infty\} \prod_{j=1}^d \text{Lip}(\pi_i^j) < \infty \quad (1.11)$$

The space of all formal Lipschitz sums (with naturally defined addition and scalar multiplication, i.e. a vector space) is denoted by $\mathcal{A}_d$. Next we need to identify formal Lipschitz sums defining the “same differential form”. The equivalence relation is given by $\sum_i (f_i, \pi_i) \sim \sum_i (g_i, \kappa_i)$ if

$$\sum_i T(f_i, \pi_i) = \sum_i T(g_i, \kappa_i) \quad \text{for every } T \in \mathcal{D}_d = \mathbb{P}_d.$$

Note that due to Theorem 1.8 the sums converge absolutely.

Potential differential forms

Definition 1.10. We define the *potential norm* on $\mathcal{A}_d / \sim$ via

$$\|\omega\|_{\mathbb{P}^d} = \|\omega\|_{\mathbb{P}^d(Q)} = \inf\{s(\sum_i (f_i, \pi_i)) : \sum_i (f_i, \pi_i) \in \omega\} \quad \text{for } \omega \in \mathcal{A}_d / \sim.$$

The set $\mathbb{P}^d(Q) = \mathbb{P}^d = \mathcal{A}_d / \sim$, equipped with the potential norm, is called *the space of (top dimensional) potential (differential) forms*¹. Its elements are called *(top dimensional) potential (differential) forms*. The map

$$\text{Em}: \mathbb{P}^d(Q) \rightarrow L^\infty(Q),$$

given by $\text{Em}([\sum_i (f_i, \pi_i)]) = \sum_i f_i \det D\pi_i$ is called *the canonical embedding (for potential forms)*.

¹If one is familiar with the basic theory of tensor products, it can be easily verified, that the space $\mathbb{P}^d$ may alternatively be obtained from the tensor product, say $\mathcal{T} = \text{Lip}_b(Q) \otimes \bigotimes_{j=1}^d \text{Lip}(Q)$. This is due to the fact that the equivalence relation we define has strictly larger equivalence classes than the equivalence relation used to define the tensor product. Thus $\mathbb{P}^d = \overline{\mathcal{T}} / \sim$

Remark 1.11. Somewhat confusingly, we have not actually proved that the canonical embedding is an embedding. I do not know how to show this, so it is left open (I will still call it an embedding though). The point is to prove that if one has two formal Lipschitz sums $\sum_i(f_i, \pi_i)$ and $\sum_i(g_i, \kappa_i)$ such that

$$\sum_i f_i \det D\pi_i = \sum_i g_i \det D\kappa_i, \quad (1.12)$$

then it also holds that $\sum_i(f_i, \pi_i) \sim \sum_i(g_i, \kappa_i)$. This would be true, for example, if $\text{Em}_*(L^1)$ were dense in $\mathbb{P}_d$.

Note that there is the natural *duality pairing* of potential currents and potential forms:

$$\langle T, \omega \rangle = \sum_i T(f_i, \pi_i) \quad \text{where } \sum_i (f_i, \pi_i) \in \omega. \quad (1.13)$$

Conjecture 1.12 (“Metric Wolffe’s theorem”). $\mathbb{P}^d = \mathbb{P}_d^*$. *Formally, to every element $\Phi \in \mathbb{P}_d^*$ there exists a potential form $\omega \in \mathbb{P}_d$ such that*

$$\langle T, \omega \rangle = \Phi(T) \quad \text{for every } T \in \mathbb{P}_d. \quad (1.14)$$

If the conjecture were true, we would be almost done with the “abstract” nonsense part. Since we cannot prove it we need to develop some more theory to actually pass to the dual problem concerning Jacobians. What we are basically trying to do is pass to the dual problem *without* identifying the dual space. This is possible but involved.

Let $I: \mathbb{P}^d \rightarrow \mathbb{P}_d^*$ be given by $I(\omega)(T) = \langle T, \omega \rangle$. Then I is surjective with operator norm at most 1. I is indicated in the bottom part of the diagram 1. Note that the bottom part of the diagram commutes.

Proposition 1.13. *For a potential current T ,*

$$\|T\|_{\mathbb{P}_d} = \sup_{\omega \in B_{\mathbb{P}_d}} \langle T, \omega \rangle. \quad (1.15)$$

It is not known whether also $\|\omega\|_{\mathbb{P}^d} = \sup_{T \in B_{\mathbb{P}_d}} \langle T, \omega \rangle$, cf. Conjecture 1.12.

1.4 Strong non-surjectivity

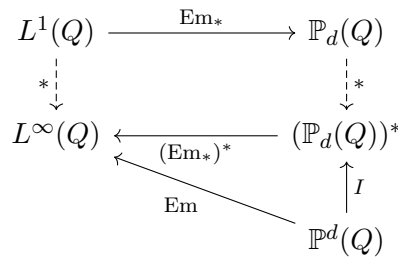


Figure 1: A diagram indicating the relationship between function spaces and potential spaces. The dashed arrows only indicate direction of duality. The full arrows correspond to operators. The bottom part of the diagram commutes.

We are in the following situation. We have three Banach spaces $Y = L^1$, $Z = \mathbb{P}_d$ and $X = \mathbb{P}^d$. Our goal is to show that $\text{Em}_*: Y \rightarrow Z$ is *not* surjective. By basic functional analysis, if $X = Z^*$, it would suffice to show that $\text{Em} = (\text{Em}_*)^*: X (= Z^*) \rightarrow Y^*$ is not surjective. Since $Y^* = L^\infty$, this is trivially the same as requiring that the PDE:

$$\sum_i f_i \det D\pi_i = \rho \quad \text{on } Q, \quad (1.16)$$

{E:PJE1}

have a solution for every right hand side $\rho \in L^\infty$. Of course by a solution we mean that there exists a formal Lipschitz sum satisfying the equality a.e.

Since we are not able to prove that $X = Z^*$ (cf. Conjecture 1.12), we have to do some more work.

{D:Lip-reg}

Definition 1.14. We say that the prescribed Jacobian equation has *strong Lipschitz non-regularity* if a sequence of $L^1(Q)$ functions u_n , $n \in \mathbb{N}$ can be found satisfying the two following inequalities simultaneously:

$$(i) \text{ for each } n \in \mathbb{N}, \|u_n\|_{L^1(Q)} \geq 1$$

$$(ii) \text{ for each } n \in \mathbb{N},$$

$$\sup \left\{ \left| \int_Q u_n f \det D\pi \, d\mathcal{L}^d \right| : \max\{\text{Lip}(f), \|f\|_\infty\} \prod_{j=1}^d \text{Lip}(\pi^j) \leq 1 \right\} \leq \frac{1}{n}.$$

{Enum:strong1}

{Enum:strong2}

Remark 1.15. Strong Lipschitz non-regularity is stronger than just “Lipschitz non-regularity”². In other words strong Lipschitz non-regularity implies that there exists some $\rho \in L^\infty$ for which the equation (1.16) does not have a solution. This is due to the open mapping theorem. If there existed a solution for every $\rho \in L^\infty$ then the operator $\text{Em} : \mathbb{P}^d \rightarrow L^\infty$ would be surjective. Then there would exist some open ball B in L^∞ containing 0 and wholly contained in

$$\left\{ \sum_i f_i \det D\pi_i : s\left(\sum_i (f_i, \pi_i)\right) \leq 1 \right\} = \overline{\text{conv}}\{f \det D\pi : s(f, \pi) \leq 1\}. \quad (1.17)$$

Thus there would be some $\varepsilon > 0$ such that for every $u \in L^1$.

$$\sup \left\{ \left| \int_Q u f \det D\pi \, d\mathcal{L}^d \right| : \max\{\text{Lip}(f), \|f\|_\infty\} \prod_{j=1}^d \text{Lip}(\pi^j) \leq 1 \right\} \\ \stackrel{\text{lin.,conv.}}{=} \sup \left\{ \left| \int_Q u \rho \, d\mathcal{L}^d \right| : \rho \in \overline{\text{conv}}\{f \det D\pi : s(f, \pi) \leq 1\} \right\} \geq \varepsilon \|u\|_{L^1}. \quad (1.18)$$

This means either (i) or (ii) must fail for any sequence of u_n .

Theorem 1.16. *The prescribed Jacobian equation has strong Lipschitz non-regularity, if and only if the range of Em_* is not closed. In particular, if the equation has strong Lipschitz non-regularity, then the embedding Em_* is not surjective.*

{T:f-a-characterisat}

Proof. By the open mapping theorem, the operator Em_* has closed range if and only if it is an isomorphism into $\mathbb{P}_d(Q)$. This is the case, if and only if there exists a constant C such that, whenever $u_n \in L^1(Q)$ satisfies (i) from Definition 1.14, then

$$\|\text{Em}_*(u_n)\|_{\mathbb{P}_d} \geq C. \quad (1.19)$$

{E:est-from-}

Indeed, C corresponds to the operator norm of the inverse of Em_* . We calculate

$$\begin{aligned} \|\text{Em}_*(u_n)\|_{\mathbb{P}_d} &= \sup \left\{ \left| \text{Em}_*(u_n)(f, \pi^1, \dots, \pi^d) \right| : \|f\|_L \prod_{j=1}^d \text{Lip}(\pi^j) \leq 1 \right\} \\ &= \sup \left\{ \left| \int_Q u_n f \det D\pi \, d\mathcal{L}^d \right| : \|f\|_L \prod_{j=1}^d \text{Lip}(\pi^j) \leq 1 \right\}. \end{aligned}$$

²This is called linearized Lipschitz non-regularity in [5]

Therefore, we see that if a constant C satisfying (1.19) exists, then the sequence u_n does not satisfy (ii) of Definition 1.14. On the other hand, if a sequence u_n can be found such that both (i) and (ii) of Definition 1.14 hold, no such constant C can exist.

The final statement follows by Lemma 1.9. $\square$

Definition 1.17 (Strong non-surjectivity). Let X and Y be Banach spaces. We say that a linear operator $T: X \rightarrow Y^*$ is *strongly non-surjective* if for every $n \in \mathbb{N}$, there exists $\rho_n \in \frac{1}{n}B_{Y^*} \cap T(X)$ and a weak* open neighbourhood U_n of ρ_n such that

$$U_n \cap T(B_X) = \emptyset.$$

Theorem 1.18 (Predual separation theorem). *Let X and Y be Banach spaces and $T: X \rightarrow Y^*$ a bounded linear operator, which is additionally strongly non-surjective. Then there exists a sequence $v_n \in Y$ such that*

- (i) $\|v_n\|_Y \geq 1$,
- (ii) $\sup_{x \in B_X} |\langle T(x), v_n \rangle| \leq \frac{1}{n}$.

Proof. For details, see the paper. Here we will just indicate the basic idea. Firstly, we recall that the weak* continuous elements of Y^{**} are precisely those that coincide with the canonical images of elements of Y . In other words, any weak* continuous linear map $\varphi: Y^* \rightarrow \mathbb{R}$ admits some $y \in Y$ such that $\varphi(\eta) = \eta(y)$ for every $\eta \in Y^*$.

Next we use Hahn-Banach to strictly separate U_n from $T(B_X)$ with a weak* continuous linear functional, say u_n which is possible due to the definition of strong non-surjectivity. Now v_n is essentially obtained by normalizing u_n and looking at what “strict separation” really means. By what we wrote above v_n and u_n are both in Y . $\square$

Theorem 1.19. *Suppose $\text{Em}: \mathbb{P}^d \rightarrow L^\infty$ is strongly non-surjective. Then $\text{Em}_*: L^1 \rightarrow \mathbb{P}_d$ is non-surjective.*

Proof. Follows by the combination of Theorem 1.16 and Theorem 1.18 applied to $X = \mathbb{P}^d$ and $Y = L^1$. $\square$

2 Strong non-regularity of the Jacobian equation

$$\begin{array}{ccc}
 L^1(Q) & \xrightarrow{\text{Em}_*} & \mathbb{P}_d(Q) \\
 \downarrow * & & \downarrow * \\
 L^\infty(Q) & \xleftarrow{(\text{Em}_*)^*} & (\mathbb{P}_d(Q))^* \\
 & \nwarrow \text{Em} & \uparrow I \\
 & & \mathbb{P}^d(Q)
 \end{array}$$

Where we left off We recall that $\mathbb{P}_d$ is the space of potential(=metric) d -currents on the unit cube $Q = [0, 1^d]$, while $\mathbb{P}^d$ stands for the space of potential differential forms - equivalence classes of formal Lipschitz sums

$$\sum_i (f_i, \pi_i), \tag{2.1}$$

which is in natural duality with $\mathbb{P}_d$ via

$$\langle T, [\sum_i (f_i, \pi_i)] \rangle = \sum_i T(f_i, \pi_i). \tag{2.2}$$

Our goal is to show that the indicated operator Em_* is not surjective. We wish to achieve this by studying instead the operator Em . This is complicated by the fact that we are not able to prove that $\mathbb{P}^d$ is the genuine dual of $\mathbb{P}_d$ and that Em is the genuine dual of Em_* .

We recall that

$$\text{Em}([\sum_i (f_i, \pi_i)]) = \sum_i f_i \det D\pi_i. \quad (2.3)$$

The property of this operator which guarantees that Em_* is non-surjective is called *strong Lipschitz non-regularity of the prescribed Jacobian equation*. This means, by definition, that there exists a sequence $u_n \in L^1$ such that both

$$\|u_n\|_{L^1} \geq 1$$

and

$$\begin{aligned} 0 &\leftarrow \sup \left\{ \left| \int_Q u_n f \det D\pi \, d\mathcal{L}^d \right| : \max\{\text{Lip}(f), \|f\|_\infty\} \prod_{j=1}^d \text{Lip}(\pi^j) \leq 1 \right\} \\ &\stackrel{\text{lin., conv.}}{=} \sup \left\{ \left| \int_Q u_n \rho \, d\mathcal{L}^d \right| : \rho \in \overline{\text{conv}}\{f \det D\pi : s(f, \pi) \leq 1\} \right\} \\ &\stackrel{\text{def.}}{=} \sup_{\|\omega\|_{\mathbb{P}^d} \leq 1} \langle \text{Em}_*(u_n), \omega \rangle \stackrel{P1.13}{=} \|\text{Em}_* u_n\|_{\mathbb{P}_d} \rightarrow 0. \end{aligned} \quad (2.4)$$

Indeed, the combination of these two properties reads as the following fact: the L^1 -norm and the $\mathbb{P}_d$ norm are not comparable (when Em_* is viewed as identity). Thus, the open mapping theorem dictates that Em_* does not have closed range and therefore is not surjective.

We follow up by introducing a general principle which allows one to get such strong non-regularity statements. We present the result in general terms. Thus, we fix two Banach spaces X and Y (in our application X is $\mathbb{P}^d$ and Y is L^1).

Definition 2.1. We say that a linear operator $T: X \rightarrow Y^*$ is *strongly non-surjective* if to each $n \in \mathbb{N}$ there is an element $\rho_n \in T(X) \cap \frac{1}{n}B_{Y^*}$ and a weak* open set $U_n \ni \rho_n$ such that

$$U_n \cap T(B_X) = \emptyset. \quad (2.5)$$

Remark 2.2. If $X = Z^*$ for some Banach space Z and $T(X)$ is weak* dense in Y^* , then T is non-surjective if and only if it is strongly non-surjective. Thus, this notion is only really useful when we are not able to verify that X is a dual Banach space. There exist non-surjective operators which are not strongly non-surjective.

Theorem 2.3. Suppose $T: X \rightarrow Y^*$ is strongly non-surjective. Then there exists a sequence $u_n \in Y$ such that both

- (i) $\|u_n\|_Y \geq 1$,
- (ii) $\sup_{x \in B_X} \langle T(x), u_n \rangle \rightarrow 0$.

Proof. A slightly involved application of the Hahn-banach separation theorem. We separate the set U_n from $T(B_X)$ (possibly after replacing the neighbourhood U_n with a convex one). The separation may be done inside Y^* with a weak* continuous functional. This functional is therefore realized by an element of the predual $v_n \in Y$. Rescaled versions of v_n provide the necessary u_n . $\square$

Theorem 2.4. Suppose the prescribed Jacobian equation has the following property. To each $n \in \mathbb{N}$ there exists a smooth map ρ_n in $2B_{L^\infty}$ together with a weak* open neighbourhood U_n such that for every $\rho \in U_n$ the equality

$$\sum_i f_i \det D\pi_i = \rho \quad \text{a.e. in } [0, 1]^d, \quad (2.6)$$

for a Lipschitz formal sum $\sum_i (f_i, \pi_i)$ implies

$$s(\sum_i (f_i, \pi_i)) \geq n. \quad (2.7)$$

Then Em_* is not surjective and hence the Flat Chain Conjecture of Lang fails.

Proof. The property, by definition, implies that Em is strongly non-surjective. Thus, the previous theorem entails strong Lipschitz non-regularity of the PJE. Finally, this implies, by the discussion above, that Em_* does not have closed range, which in turn finally implies that the said operator is non-surjective. $\square$

Remark 2.5. The reason ρ_n is required to be smooth is to guarantee that $\rho_n \in \text{Em}(\mathbb{P}^d)$, cf. the requirement that $\rho_n \in T(X)$ in Definition 2.1.

2.1 How to deal with the weak* open neighbourhood

Below we will indicate how to obtain the ρ_n but we will first indicate how the weak* open neighbourhood is obtained. To each $n \in \mathbb{N}$ we will construct a finite (although an enormously huge) collection of subcubes $Q_i \subset [0, 1]^d$. Our ρ_n will be constructed in such a way that its averages over the cubes Q_i are (up to a tiny error) prescribed. Say that we have something like

$$\frac{1}{\mathcal{H}^d(Q_i)} \int_{Q_i} \rho_n \in (a_i - \varepsilon_i, a_i + \varepsilon_i), \quad (2.8)$$

where a_i are some real numbers and ε_i are some positive numbers. We then prove, using only this set of average estimates that ρ_n has the required property - namely that any FLS solving the PDE has the s -norm larger than n .

This means the same can be said about *every* ρ satisfying the same set of average estimates. Thus, the weak* open neighbourhood is obtained. Indeed the set

$$\bigcap_i \left\{ \rho : \frac{1}{\mathcal{H}^d(Q_i)} \int_{Q_i} \rho \in (a_i - \varepsilon_i, a_i + \varepsilon_i) \right\} \quad (2.9)$$

is, by definition, weak* open since the set of indices over which we are intersecting is finite.

2.2 Constructing the ρ_n

This is based on earlier constructions of Burago and Kleiner [2] and Dymond, Kaluža and Kopecká [3]. We first consider the simplified equation

$$\det D\pi = \rho \quad \text{a.e. in } [0, 1]^d, \quad (2.10)$$

and given some $L > 0$ we will indicate how one can construct a measurable map $\rho: [0, 1]^d \rightarrow \{1, 2\}$ such that any Lipschitz map π solving the PDE above must have Lipschitz constant at most L (this is what Burago and Kleiner do). We remark that by a kind of normalisation procedure, one may infer from this that any such map must also satisfy $\prod_{j=1}^d \text{Lip}(\pi^j) \geq (L/d)^d$.

The very general idea is to find a property that determinants of Lipschitz maps must have that a general L^∞ map need not have. This is obtained using two ingredients - a Stokes theorem for determinants and a quantitative differentiation theorem for Lipschitz maps (in 1D).

Lemma 2.6. *For every $d \in \mathbb{N}$, $d \geq 2$, there exists a constant $c_d \in (0, \infty)$ such that the following holds. For every cube $Q \subset \mathbb{R}^d$ of side length $r > 0$, and every pair of Lipschitz maps $\pi, \kappa: Q \rightarrow \mathbb{R}^d$, it holds that*

$$\left| \int_Q \det D\pi - \int_Q \det D\kappa \right| \leq c_d L^{d-1} r^{d-1} \|\pi - \kappa\|_{\ell^\infty(\partial Q)}, \quad (2.11)$$

where L is the maximum of the Lipschitz constants of the functions $\pi^1, \dots, \pi^d, \kappa^1, \dots, \kappa^d$.

{L:Average-c

{E:principal
estimate}

Proof. Some multilinear algebra together with the Stokes theorem:

$$\int_Q \det D\pi d\mathcal{H}^d = \int_{\partial Q} \pi_1 \det D(\pi_2, \dots, \pi_d) d\mathcal{H}^{d-1} \quad (2.12)$$

{E:principal estimate}

□

Proposition 2.7 (Qualitative result). *Suppose $g: [0, 1] \rightarrow \mathbb{R}$ is 1-Lipschitz and $g(0) = 0$ and $g(1) = 1$. Then g is the identity.*

Proof. $g' \leq 1$ since g is 1-Lipschitz. But also $\int_0^1 g' = g(1) - g(0) = 1$ by the fundamental theorem of calculus. Thus, $g' = 1$ a.e. and so $g = \text{Id}$. □

Proposition 2.8 (First quantitative result). *To each $\delta > 0$ there exists $\varepsilon > 0$, such that if $g: [0, 1] \rightarrow \mathbb{R}$ is 1-Lipschitz and $|g(0) - 0| < \varepsilon$, $|g(1) - 1| < \varepsilon$, then*

$$\mathcal{H}^1\{t \in [0, 1] : |g'(t) - 1| > \delta\} \leq 1 - \delta.$$

In the following, “lattice points” refer to points in $[0, 1]^c$ of the form $\frac{1}{n}c$ where $n \in \{1, \dots, N\}$. “Sublattice points” refer to points of the form $\frac{1}{nm}c$ where n is as above and $m \in \{1, \dots, M\}$.

Theorem 2.9 (Dymond, Kaluža, Kopecká; earlier form due to Burago, Kleiner). *Let $L \geq 0$ and $\varepsilon > 0$. Then there exists $M \in \mathbb{N}$ and $\varphi > 0$ such that the following holds. For every $c > 0$, $n \in \mathbb{N}$, $h: [0, c] \rightarrow \mathbb{R}^n$ L -Lipschitz and every $N \geq 2$, one of the following statements is true.*

(i) *The set of lattice points x , where*

$$\|h(x + \frac{c}{N}) - h(x) - \frac{1}{N}(h(c) - h(0))\| \leq \frac{c\varepsilon}{N},$$

takes up at least $(1 - \varepsilon)$ proportion of all lattice points.

(ii) *There exists a sublattice point z such that*

$$\frac{\|h(z) - h(z + \frac{c}{NM})\|}{\frac{c}{NM}} \geq (1 + \varphi) \frac{\|h(0) - h(c)\|}{c}.$$

Finally, we need a version of this theorem for maps from very thin rectangles of dimension d . Here sublattice points are the points on the $M \times M$ regular lattice of each subcube of side length $\frac{1}{N}c$ of the rectangle $[0, c] \times [0, \frac{1}{N}c]^d$. Those cubes are called lattice cubes. See figure 3 where the dots are sublattice points and the cubes are the lattice cubes.

Theorem 2.10 (Dymond, Kaluža, Kopecká; earlier form due to Burago, Kleiner). *Let $L \geq 0$ and $\varepsilon > 0$. Then there exists $M \in \mathbb{N}$, $\varphi > 0$ and $N_0 \in \mathbb{N}$ such that the following holds. For every $N \geq N_0$, $c > 0$, $n \in \mathbb{N}$ and $g: [0, c] \times [0, \frac{c}{N}]^{d-1}$, L -biLipschitz one of the following statements is true.*

(i) *The set of lattice cubes Q such that **for all** $x \in Q$:*

$$\|h(x + \frac{c}{N}e_1) - h(x) - \frac{1}{N}(h(ce_1) - h(0))\| \leq \frac{c\varepsilon}{N},$$

takes up at least $(1 - \varepsilon)$ proportion of all lattice cubes.

(ii) *There exists a sublattice point z such that*

$$\frac{\|h(z) - h(z + \frac{c}{NM}e_1)\|}{\frac{c}{NM}} \geq (1 + \varphi) \frac{\|h(0) - h(ce_1)\|}{c}.$$

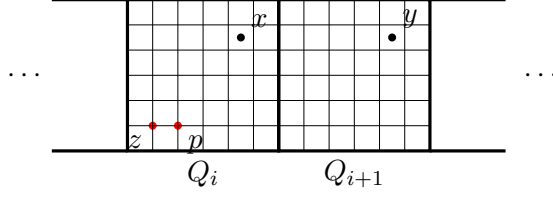


Figure 2: b

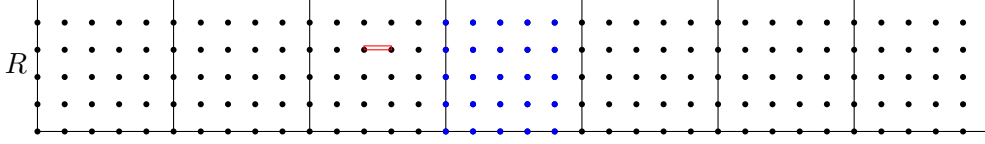


Figure 3: a

{fig:adjacent
squares-
dichotomy}

{fig:lattice}

Putting aside a number of significant technical difficulties, one might have the following idea. Suppose we start with some right side ρ to the equation and consider an L -Lipschitz solution π . Suppose we have an apriori estimate on the distance $|\pi(0, \dots, 0) - \pi(1, 0, \dots, 0)| \geq \delta$. Then we either have a pair of adjacent lattice cubes Q, Q' in the rectangle $[0, 1] \times [0, \frac{1}{N}]^d$ such that π behaves almost identically on Q and Q' (statement (i) holds) or we have a pair of adjacent sublattice points z, p such that $|\pi(z) - \pi(p)| \geq (1 + \varphi)\delta|z - p|$. In the latter case, we look at the tiny rectangle between z, p (see the red rectangle in figure 3). We continue this process until we get the statement (i) to hold. This must happen since if statement (i) does not occur k times we will have a pair of lattice points z, p mapped farther away than $(1 + \varphi)^k \delta|z - p|$ eventually outgrowing the Lipschitz constant L , which may not happen since π is L -Lipschitz.

So we may affirm that after a number of steps depending only on L (and δ , which is a difficulty that can be resolved) we find a pair of adjacent cubes Q, Q' where π behaves almost the same on Q as it does on Q' . However, this in combination with the Stokes theorem (Lemma 2.6) gives

$$|\int_Q \det D\pi - \int_{Q'} \det D\pi| \lesssim \varepsilon r^{d-1} r \implies |\frac{1}{\mathcal{H}^d(Q)} \int_Q \det D\pi - \frac{1}{\mathcal{H}^d(Q')} \int_{Q'} \det D\pi| \leq \varepsilon$$

where r is the sidelength of Q . Thus, we see that we only need to construct ρ such that the difference of the averages of ρ on all pairs of adjacent lattice cubes up to some predetermined depth (depending on L) is always large (say around $1/2$).

Remark 2.11. By Rademacher and Egorov, for any fixed Lipschitz map π we may always find a pair of some adjacent cubes (in fact, in some sense, this will be satisfied by the majority of tiny enough adjacent cubes) such that the averages of $\det D\pi$ do not differ much between the cubes. This does not suffice as the same can be said about any fixed right side ρ (by Egorov). The point is that we may *predetermine* a finite set of cubes on which this phenomenon *must* be observed for *every* L -Lipschitz map π .

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